Foundations of Gaussian process

[Gaussian process]

$$f(x) \sim \mathcal{GP}(m(x),\, k(x, x'))$$

1.0 Content Block

A Gaussian process can be used as a prior probability distribution over functions in Bayesian inference. Given any set of N points in the desired domain of the functions, take a multivariate Gaussian whose covariance matrix parameter is the Gram matrix of those N points with some desired kernel, and sample from that Gaussian. For solution of the multi-output prediction problem, Gaussian process regression for vector-valued function was developed. In this method, a 'big' covariance is constructed, which describes the correlations between all the input and output variables taken in N points in the desired domain. This approach was elaborated in detail for the matrix-valued Gaussian processes and generalised to processes with 'heavier tails' like Student-t processes. Inference of continuous values with a Gaussian process prior is known as Gaussian process regression, or kriging; extending Gaussian process regression to multiple target variables is known as cokriging. Gaussian processes are thus useful as a powerful non-linear multivariate interpolation tool. Kriging is also used to extend Gaussian process in the case of mixed integer inputs. Gaussian processes are also commonly used to tackle numerical analysis problems such as numerical integration, solving differential equations, or optimisation in the field of probabilistic numerics. Gaussian processes can also be used in the context of mixture of experts models, for example. The underlying rationale of such a learning framework consists in the assumption that a given mapping cannot be well captured by a single Gaussian process model. Instead, the observation space is divided into subsets, each of which is characterized by a different mapping function; each of these is learned via a different Gaussian process component in the postulated mixture.

2.0 Content Block

where ξ 1 , η 1 , ξ 2 , η 2 , … {\displaystyle \xi _{1},\eta _{1},\xi _{2},\eta _{2},\dots } are independent random variables with standard normal distribution; frequencies 0 < λ 1 < λ 2 < … {\displaystyle 0<\lambda _{1}<\lambda _{2}<\dots } are a fast growing sequence; and coefficients c n > 0 {\displaystyle c_{n}>0} satisfy ∑ n c n < ∞ . {\textstyle \sum _{n}c_{n}<\infty .} The latter relation implies

3.0 Content Block

Some history. Sufficiency was announced by Xavier Fernique in 1964, but the first proof was published by Richard M. Dudley in 1967. Necessity was proved by Michael B. Marcus and Lawrence Shepp in 1970. There exist sample continuous processes X {\displaystyle X} such that I (σ) = ∞ ; {\displaystyle I(\sigma)=\infty ;} they violate condition (*). An example found by Marcus and Shepp is a random lacunary Fourier series

4.0 Content Block

E [X t X t + h] = ∑ n = 1 ∞ c n 2 cos ⁡ λ n h {\displaystyle {\mathbb {E} }[X_{t}X_{t+h}]=\sum _{n=1}^{\infty }c_{n}^{2}\cos \lambda _{n}h}

5.0 Content Block

var ■ [X (t)] = E [| X (t) − E ■ [X (t)] | 2] < ∞ for all t ∈ T . {\displaystyle \operatorname {var} [X(t)]={\mathbb {E} }\left[\left|X(t)-\operatorname {E} [X(t)]\right|^{2}\right]<\infty \quad {\text{for all }}t\in T.}

6.0 Content Block

these two integrals being equal according to integration by substitution h = e − x 2 , {\textstyle h=e^{-x^{2}},} x = log ■ (1 / h) . {\textstyle x={\sqrt {\log(1/h)}}.} The first integrand need not be bounded as h → 0 + , {\displaystyle h\to 0+,} thus the integral may converge (I (σ) < ∞ {\displaystyle I(\sigma)<\infty }) or diverge (I (σ) = ∞ {\displaystyle I(\sigma)=\infty }). Taking for example σ (e − x 2) = 1 x a {\textstyle \sigma (e^{-x^{2}})={\tfrac {1}{x^{a}}}} for large x , {\displaystyle x,} that is, σ (h) = (log ■ (1 / h)) − a / 2 {\textstyle \sigma (h)=(\log(1/h))^{-a/2}} for small h , {\displaystyle h,} one obtains I (σ) < ∞ {\displaystyle I(\sigma)<\infty } when a > 1 , {\displaystyle a>1,} and I (σ) = ∞ {\displaystyle I(\sigma)=\infty } when 0 < a ≤ 1. {\displaystyle 0<a\leq 1.}

7.0 Content Block

Brownian motion as the integral of Gaussian processes A Wiener process (also known as Brownian motion) is the integral of a white noise generalized Gaussian process. It is not stationary, but it has stationary increments. The Ornstein–Uhlenbeck process is a stationary Gaussian process. The Brownian bridge is (like the Ornstein–Uhlenbeck process) an example of a Gaussian process whose increments are not independent. The fractional Brownian motion is a Gaussian process whose covariance function is a generalisation of that of the Wiener process.

8.0 Content Block

Linearly constrained Gaussian processes For many applications of interest some pre-existing knowledge about the system at hand is already given. Consider e.g. the case where the output of the Gaussian process corresponds to a magnetic field; here, the real magnetic field is bound by Maxwell's equations and a way to incorporate this constraint into the Gaussian process formalism would be desirable as this would likely improve the accuracy of the algorithm. A method on how to incorporate linear constraints into Gaussian processes already exists: Consider the (vector valued) output function f (x) {\displaystyle f(x)} which is known to obey the linear constraint (i.e. F X {\displaystyle {\mathcal {F}}_{X}} is a linear operator)

9.0 Content Block

Periodic: K P (x , x ′) = exp ■ (− 2 ■ 2 sin 2 ■ (d / 2)) {\displaystyle K_{\operatorname {P} }(x,x')=\exp \left(-{\tfrac {2}{\ell ^{2}}}\sin ^{2}(d/2)\right)}

10.0 Content Block

f (x) = G X g ∼ G P (G X μ g , G X K g G X ′ T) . {\displaystyle f(x)={\mathcal {G}}_{X}g\sim {\mathcal {GP}}({\mathcal {G}}_{X}\mu _{g},{\mathcal {G}}_{X}K_{g}{\mathcal {G}}_{X'}^{\mathsf {T}}).}

11.0 Content Block

Given G X {\displaystyle {\mathcal {G}}_{X}} and using the fact that Gaussian processes are closed under linear transformations, the Gaussian process for f {\displaystyle f} obeying constraint F X {\displaystyle {\mathcal {F}}_{X}} becomes

12.0 Content Block

(the right-hand side does not depend on t {\displaystyle t} due to stationarity). Continuity of X {\displaystyle X} in probability is equivalent to continuity of σ {\displaystyle \sigma } at 0. {\displaystyle 0.} When convergence of σ (h) {\displaystyle \sigma (h)} to 0 {\displaystyle 0} (as h → 0 {\displaystyle h\to 0}) is too slow, sample continuity of X {\displaystyle X} may fail. Convergence of the following integrals matters: